Project work

Deadline: Sun 16.4. 23:59

1. Form groups of 2-6 students. Larger group -> broader outcomes
2. Pick one of the approaches:
    1. Using real world data, formulate a scientific question you can answer using some of the metrics learned on the course
    2. Use Kuramoto time series data to study avalanche dynamics, metastable states, or something else relevant
    3. Run a behavioural experiment (code your own or use existing implementations) and explore behavioural scaling laws
    4. Formulate and answer a scientific question with modeling

1. Real-world data

Where do I find data?

1. MNE example datasets: https://mne.tools/stable/overview/datasets_index.html
2. Some other sources listed here: https://www.fieldtriptoolbox.org/faq/open_data/
3. Many other places around the internet
4. We can provide some data (single subject data)
5. You can use your own data

Metrics include but not limited to:

- Synchrony (PLV, PLF)
- Criticality (DFA)
- Evoked, induced responses
- ...

2. Kuramoto time series

Where do I find data?

- Data provided by us, or you can create your own

Avalanche dynamics

Metastable states

Invent a metric or go through literature!

3. Code your own behavioural experiment

Where do I find data?

- Behavioural data from the task you code, or find online
  - Reaction times, commission errors, etc.
  - Collect data from subjects (could be even group members)

Behavioural scaling laws

4. Modelling

Where do I find data?

- Example connectome available

Formulate your scientific question

Resources

- GPU resources:
  - Google Colab
  - Paniikki computer lab in Aalto – All computers have GPU
  - Remote connect to Aalto work stations -> pick GPU workstation

- Computers around Aalto (e.g. Maari computers are fast and have RAM)

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3. Write a report using the following structure:

Introduction

- Background
- Problem statement & knowledge gap, suitable for course scope
- Question addressed in this project
- Hypothesis (about the expected result)
- Aim
- Approach

Methods

Results

Discussion

References

Project scoring – 50% code

Preprocessing – 2pts

- Preprocessing done with proper and justified methods – 2pts
- Preprocessing done okay'ish but some necessary step is missing – 1pt
- Preprocessing not done or without proper and justified methods – 0pts

Methods – 2pts

- Suitable methods for the question – 1pt
- Correct implementation of the methods – 1pt

Visualization & analyses – 3 pts

- Proper visualization, including interim results where applicable – 1pt
- Analysis tools are suited for the task – 1pt
- Statistics included; research backed by it – 1pt

Project scoring – 50% report

Introduction/background – 2 pts

- ~50% of them – 1pt
- All of them – 2pts

Methods – 1 pt

- Every method presented in an understandable and reproducible way – 1pt

Results – 2 pts

- Statistics backed results – 1pt
- Results answer the question, the test hypothesis and achieve the aim – 1pt

Discussion – 2 pts

- Concise summary of the results, must mirror the aim, hypothesis and problem introduced through the results – 1pt
- Caveats, limitations, future directions – 1pt